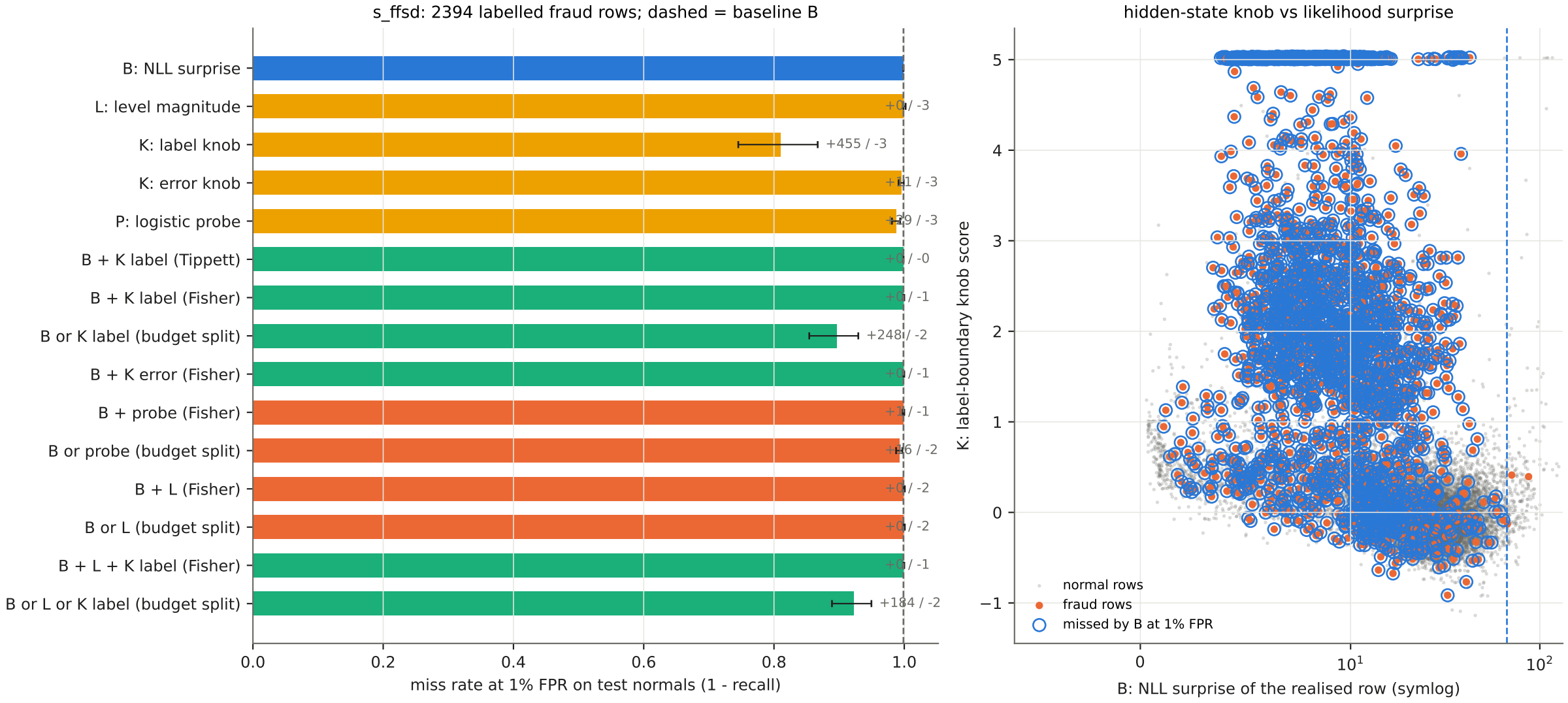

s_ffsd: 2394 labelled fraud rows; dashed = baseline B
B: NLL surprise
L: level magnitude
K: label knob
K: error knob
P: logistic probe
B + K label (Tippett)
B + K label (Fisher)
B or K label (budget split)
B + K error (Fisher)
B + probe (Fisher)
B or probe (budget split)
B + L (Fisher)
B or L (budget split)
B + L + K label (Fisher)
B or L or K label (budget split)
+455 / -3
+248 / -2
+184 / -2
0.0
0.2
0.4
0.6
0.8
1.0
miss rate at 1% FPR on test normals (1 - recall)
hidden-state knob vs likelihood surprise
K: label-boundary knob score
normal rows
fraud rows
missed by B at 1% FPR
B: NLL surprise of the realised row (symlog)